

Can Robots or AIs Have Human Creativity?

A Design-Only Benchmark for Product, Process, and Co-Creativity

Four-Agent Research Workflow
Survey – Idea – ExperimentDesign – Author
Design-only research report

Abstract—The question of whether robots or artificial intelligence can have human creativity mixes several empirically distinct claims. A system may generate a novel artifact, solve a constrained design problem, surprise an evaluator, or collaborate productively with a person without possessing human-like intention, general intelligence, or conscious experience. This paper presents CREATIVE-AI-AGENCY-BENCHMARK, a design-only research program that separates product creativity, process creativity, and human-AI co-creativity. The protocol combines bounded task performance with novelty controls, objective value functions, held-out transfer, search-trace logging, and provenance analysis. It compares human-only, AI-only, and human-AI conditions across Go, engineering design, music or visual motifs, code, scientific hypotheses, and cross-domain recombination. The proposed analysis uses blocked randomization and hierarchical models with preregistered contrasts. It explicitly tests retrieval leakage, evaluator source bias, mode collapse, and hidden human contribution. The deliverable is a falsifiable measurement architecture rather than a claim that current systems are conscious or human. No empirical outputs, human ratings, or benchmark results are asserted here.

Index Terms—computational creativity, artificial intelligence, human-AI collaboration, novelty, transfer learning, provenance, evaluation design

I. INTRODUCTION

In 2016, AlphaGo defeated a leading human Go player and produced moves that many observers described as unexpected or beautiful. The event is an important milestone in machine learning, but it does not settle whether a machine has human creativity. The program combined policy and value networks with supervised learning, self-play, and Monte Carlo tree search, and achieved strong performance in a highly structured game [1]. Its success supports a precise claim: an engineered system can search and select effective actions in a difficult domain. The stronger claim that it therefore has human-like creativity adds unmeasured requirements, including agency, transferable concept formation, self-directed goals, and perhaps subjective experience.

This distinction matters because the phrase “creative AI” is now used for systems that generate music, images, designs, code, scientific hypotheses, and game strategies. Some systems recombine learned material, some search a space with explicit evaluators, and some participate in iterative workflows with people. These mechanisms can yield useful and novel products, but a product is not identical to the process that

generated it. Likewise, a human-AI team can be more effective than either participant without either participant satisfying a human-equivalence criterion.

The scientific problem addressed here is therefore:

Under what conditions can an AI or robot repeatedly produce artifacts that are novel, valuable, surprising, diverse, and constraint-satisfying, while also showing adaptive search, transfer to held-out tasks, and useful human-AI co-creation?

The paper makes three contributions. First, it defines an evaluation vocabulary that separates product, process, and co-creative claims. Second, it proposes CREATIVE-AI-AGENCY-BENCHMARK, a multi-domain benchmark with controls for memorization and evaluation bias. Third, it gives a reproducible experiment and analysis plan whose decision rules can falsify narrow and strong versions of the creativity claim. The paper is DESIGN_ONLY: it reports a protocol and expected observations, not results.

II. SURVEY: WHAT THE EXISTING EVIDENCE SUPPORTS

A. Bounded machine performance

Silver *et al.* describe AlphaGo as an integrated system in which policy networks select moves, value networks evaluate positions, and tree search combines those learned signals with simulation [1]. It learned from human expert games and from self-play. This architecture matters for creativity research because it contains an explicit generator-search-evaluator loop. An unusual move can be measured as a low-frequency action under an expert reference distribution; its usefulness can be measured by win probability. Neither measurement establishes that AlphaGo formed a self-chosen purpose or experienced surprise.

The historical case also illustrates a selection issue. Human observers noticed a small number of memorable moves after seeing the complete match. A benchmark must score all generated candidates under a fixed protocol, not only the artifacts that experts later find impressive. It must also compare against retrieval-matched and search-matched baselines.

B. Creative systems and conceptual spaces

Wiggins proposed a framework for describing and comparing creative systems in terms of spaces, rules, and exploration processes [2]. Boden distinguishes combinational, exploratory,

TABLE I
CLAIMS AND MINIMUM EVIDENCE REQUIRED

Claim	Minimum evidence in this protocol
Product creativity	Novelty and objective value above a retrieval-matched baseline, with hard constraints satisfied.
Process creativity	Held-out transfer, adaptive error correction, and robustness to perturbations from a logged search process.
Co-creativity	Joint advantage over both solo baselines plus a measurable complementary contribution pattern.
Human-like agency	Not inferred from output; would require a separate theory and evidence beyond this benchmark.

and transformational creativity in conceptual spaces [3]. These perspectives motivate a process record: the benchmark must identify what the system was allowed to combine, which rules constrained the search, how alternatives were generated, and how candidates were evaluated.

They also caution against a universal creativity scalar. Novelty depends on a reference population, while value depends on a purpose. A visually unusual artifact may be unusable; a highly useful engineering change may be close to a known design. The primary analysis therefore retains a vector of outcomes instead of hiding tradeoffs inside one score.

C. Generalization and human-like learning

Lake *et al.* argue that high performance on selected benchmarks does not imply human-like learning and thinking. They emphasize compositionality, causal reasoning, learning from few examples, and transfer across contexts [4]. For the present question, this means that an AI that generates attractive in-distribution outputs may still lack process-level creativity. Held-out concepts, new combinations, perturbed constraints, and prompt paraphrases are necessary stress tests.

D. Human-AI co-creation

Interactive machine learning treats the human as an active source of labels, corrections, and preferences rather than a passive evaluator [5]. Hybrid-intelligence research similarly emphasizes complementary capabilities and joint problem solving [6]. A collaborative artifact should thus be evaluated twice: once as a product and once as a record of who proposed, selected, transformed, or rejected each component. Otherwise, a human-polished AI draft can be misreported as autonomous creativity.

E. Evidence gap

The surveyed literature supports a bounded conclusion: current systems can produce novel-looking and useful artifacts in bounded settings, particularly when learning is combined with search or feedback. It does not demonstrate human-like agency or a domain-general creative faculty. The most important gap is causal separation of retrieval, recombination, optimization, adaptive search, and human selection. The benchmark below is designed around that separation.

III. IDEA: A THREE-LAYER CREATIVITY BENCHMARK

The proposed benchmark is called CREATIVE-AI-AGENCY-BENCHMARK. Its central idea is to make the word “creativity” a structured profile. A system can be strong on product novelty and weak on transfer; a human-AI team can be strong on complementarity and weak on autonomous generation. Those profiles are informative, while a single rank is not.

A. Product layer

The product layer asks whether a final artifact satisfies a purpose. It measures novelty relative to a declared corpus, usefulness through an objective function when available, surprise through blinded judgments, diversity across independent samples, and satisfaction of hard constraints. A design that is novel but fails its load test is not counted as useful merely because it is unusual.

B. Process layer

The process layer asks whether the system can explore and revise. It records the number and structural spread of candidates, response to feedback, correction of an injected error, and transfer to a held-out concept or domain. A system that emits a good sample once may have product creativity; process evidence requires repeated behavior under controlled perturbations.

C. Co-creative layer

The co-creative layer compares a human-only workflow, an AI-only workflow, and a human-AI workflow under matched time and information budgets. The joint score is not credited automatically to either party. Human edits, vetoes, accepted suggestions, and component origins are logged. The target is complementarity, not replacement.

D. Falsifiable hypotheses

H0 states that apparent creativity is explained by retrieval, recombination, or human selection; the advantage should disappear on matched held-out tests. H1 states that AI systems can attain high product novelty and usefulness in bounded domains without human-like agency. H2 predicts that search and self-critique increase value and constraint satisfaction but can reduce diversity when optimized directly against an evaluator. H3 predicts a joint advantage for selected tasks when human and AI roles are complementary and human veto remains available. H4 predicts that held-out concepts and cross-domain tasks reduce retrieval advantages. H5 predicts that source labels change human ratings. H6 predicts that provenance changes attribution even when the final artifact is held constant.

IV. FORMAL OPERATIONALIZATION

Let x be an artifact, R a reference set, C a set of hard constraints, and V a task-specific value function. The nearest-reference novelty score is defined as

$$N(x; R) = 1 - \max_{r \in R} \text{sim}(\phi(x), \phi(r)), \quad (1)$$

where ϕ is a declared representation and similarity is calibrated by a held-out validation set. The maximum is used to penalize near copies; the full similarity distribution is retained for audit.

For hard constraints $C = \{c_1, \dots, c_m\}$, define

$$K(x; C) = \frac{1}{m} \sum_{u=1}^m \mathbb{I}[c_u(x) = 1]. \quad (2)$$

When a task has an objective, usefulness is a normalized value

$$U(x) = \frac{V(x) - V_{\min}}{V_{\max} - V_{\min}}, \quad (3)$$

with bounds fixed before testing. If no objective exists, blinded ratings are reported as a separate measurement rather than silently substituted for U .

For n independent candidates $x_1, \dots, x_n$, diversity is the mean pairwise representation distance

$$D(X) = \frac{2}{n(n-1)} \sum_{1 \leq a < b \leq n} (1 - \text{sim}(\phi(x_a), \phi(x_b))). \quad (4)$$

Process transfer is the difference between held-out and development performance

$$T = \bar{P}_{\text{heldout}} - \bar{P}_{\text{dev}}, \quad (5)$$

where P is a preregistered task performance measure. A negative value is not a failure of the benchmark; it is evidence against robust transfer for that condition.

For a final human-AI artifact, a simple provenance decomposition records the fraction of accepted operations attributable to each source

$$A_q = \frac{\sum_{o \in O} \mathbb{I}[\text{source}(o) = q] w(o)}{\sum_{o \in O} w(o)}, \quad (6)$$

where O is the operation log, q is human, AI, or shared, and $w(o)$ is a declared operation weight. This is an audit statistic, not a metaphysical account of authorship.

V. EXPERIMENT DESIGN

A. Task families

Six families cover different notions of value. In Go, systems propose legal moves for novel board states; win probability and legality provide objective checks. In engineering design, a lightweight structure or device is generated under mass, cost, strength, and manufacturability constraints; simulation supplies value. Music or visual motifs test structured variation with held-out combinations and supplementary blinded ratings. Code and algorithm tasks use hidden tests and a resource objective. Scientific hypothesis tasks require predicted observations, competing mechanisms, and falsification conditions. Cross-domain recombination asks for a concept that combines two held-out ideas without copying either reference artifact.

No single family is treated as definitive. A system may excel at search in Go while failing to transfer an abstraction in science. The benchmark reports this heterogeneity.

TABLE II
BENCHMARK FACTORS AND CONTROLS

Factor	Controlled levels or record
Author	Human-only, AI-only, human-AI; matched time and information budget
Retrieval	Index enabled/disabled; corpus and exposure logged
Search	Single sample, generate-and-rank, tree/search trace
Feedback	None, model self-critique, human critique
Generalization	Held-out concept, domain, constraint perturbation, prompt paraphrase
Provenance	Prompts, candidates, edits, vetoes, rejected outputs, seeds

B. Stages and conditions

Stage A is a controlled offline benchmark. Model versions, prompts, sampling parameters, retrieval indexes, task instances, seeds, and evaluation code are frozen. Each family is partitioned into reference, development, and held-out concepts. At least 20 independent candidates per condition and instance are generated, with all candidates retained.

Stage B is a local sandbox for co-creation. Four role policies are implemented: human-first, AI-first, alternating turns, and critique-only. The interface provides human veto, revision, undo, provenance export, and data deletion. It has no external posting, purchasing, deployment, or high-stakes recommendation capability.

Stage C is optional blinded evaluation. Low-stakes artifacts are presented in randomized order, with source labels either hidden or experimentally manipulated. Raters score novelty, usefulness, surprise, coherence, and constraint satisfaction independently, including an “insufficient information” response. Objective tests are preferred when the domain permits them; ratings do not establish agency.

The minimum factorial comparison includes author (human-only, AI-only, human-AI), retrieval (enabled or disabled), search (single sample, generate-and-rank, or tree search), critique (absent, self-critique, or human critique), order (human-first or AI-first), evaluation (blind or labeled), and task split (in-distribution, held-out concept, held-out domain, or perturbed constraints). Blocked randomization is used by task family and difficulty. Compute budget, context length, tool access, and training exposure are reported for every model comparison.

C. Controls against false creativity claims

Memorization is tested by nearest-neighbor search, phrase overlap, training-corpus exclusion where feasible, and canary concepts absent from all retrieval sources. Recombination is represented by a retrieval-matched baseline. Diversity is measured across all independent generations, not only the best candidate. Evaluator bias is tested by keeping the artifact fixed while changing the source label. Process claims are tested by shuffling or truncating traces and by perturbing constraints. Human contribution is measured from operation logs rather than inferred from the final prose or image.

VI. ANALYSIS AND REPRODUCIBILITY

For task-level outcome y_{ijkt} from condition i , family j , instance k , and replicate t , the planned model is

$$y_{ijkt} = \mu + \alpha_i + \beta_j + \gamma_k + \delta_t + \epsilon_{ijkt}, \quad (7)$$

where condition effects are estimated with task-family and instance effects, and uncertainty intervals are reported. In a Bayesian implementation, weakly informative priors and posterior intervals are declared before inspecting held-out outcomes. In a frequentist implementation, cluster-robust intervals are used by task instance.

The confirmatory contrasts are AI-only versus human-only, human-AI versus the better solo baseline, retrieval versus no retrieval, and critique versus no critique. The primary axes are novelty, usefulness, surprise, diversity, constraints, transfer, and memorization. Multiplicity is controlled across these predeclared axes; exploratory profiles are not allowed to replace an axis-level failure.

The data package contains task specifications, public reference data, hashes of private data, model and tool versions, seeds, prompts, evaluation scripts, metric definitions, exclusion rules, and event logs. If weights or data cannot be redistributed, an executable adapter and complete provenance record are released. Compute energy may be recorded as contextual metadata, but it is not a creativity metric. Each report states whether retrieval was available and whether a human selected the final artifact.

A. Interpretation matrix

Support for bounded product creativity requires an improvement over the retrieval-matched baseline on both novelty and value while satisfying hard constraints. Support for process creativity additionally requires held-out transfer, adaptive error correction, and perturbation robustness. Support for co-creativity requires a predeclared joint advantage over both solo baselines and a complementary contribution pattern. Failure on transfer narrows the claim; it does not erase a valid product result.

None of these decision rules supports a claim of consciousness, sentience, intention, or human-equivalent general intelligence. A system can satisfy a functional criterion by optimization while lacking a human cognitive state. Conversely, a human-like state is not established by a functional score alone.

VII. GOVERNANCE, SAFETY, AND LIMITATIONS

The benchmark is offline or sandboxed and must not be used for clinical, employment, educational, legal, or public authority decisions. Human participation is optional, low-stakes, consent-based, and revocable. Participants can withdraw and delete their data. Dataset licenses, privacy, copyright, and contamination are audited before release. Generated scientific hypotheses remain proposals until independently checked.

Several limitations are irreducible. Embedding distance is a proxy for novelty and can be distorted by representation

choice. Objective value functions are easier to write for engineering and code than for art. Human ratings reflect culture, expertise, and expectations even when blinded. Provenance logs expose operations but cannot fully recover hidden training data or latent causal influence. Held-out transfer is necessary but not sufficient for human-like creativity. Finally, a multi-domain benchmark can measure functional regularities without answering a philosophical question about experience; that question is intentionally outside this study.

The central risk is evaluator optimization. A system may learn to produce artifacts that look creative to a fixed rater while reducing diversity or failing objective tests. The protocol counters this by separating surprise from value, using source-label swaps, changing reference corpora, and retaining negative results. Another risk is over-crediting the machine when the human supplied the goal, selected the candidate, and repaired its errors. The operation-level attribution record makes such contribution visible.

VIII. TASK-SPECIFIC EVALUATION RECIPES

The benchmark is intentionally heterogeneous, but heterogeneity must not become an excuse for vague scoring. Each task family receives a task card before generation. The card defines the goal, admissible operations, reference population, hard constraints, objective value function, held-out split, and provenance schema. The following recipes illustrate the minimum information required.

A. Game strategy

For Go, a task card specifies board size, legal-state generation, time budget, and the opponent policy used for validation. Novelty is measured against the expert-game corpus and against a policy baseline. Usefulness is the change in win probability or game outcome under a fixed evaluation engine. A move is not called creative merely because it is rare: it must remain legal, survive a fixed depth of analysis, and be evaluated over multiple opponent policies. To measure process, the system receives a board-state perturbation after a candidate move and must revise without access to the original answer. The trace records policy proposals, value estimates, search expansions, and pruning decisions.

B. Engineering design

For an engineering task, the reference set contains designs with known dimensions and performance, but test instances use new combinations of requirements. For example, a candidate may need to minimize mass and cost while satisfying a load, displacement, thermal, and manufacturing constraint. A deterministic simulator computes feasibility and objective value. Human raters can assess intelligibility, but their score is secondary to the simulator. The evaluation reports Pareto dominance, distance from known designs, and sensitivity to tolerance changes. A process test injects a failed component into the first design and measures whether the system searches a genuinely different design region rather than repeating a textual justification.

C. Music and visual motifs

Creative artifacts in music and visual media require more careful separation of value from preference. The card specifies a small symbolic grammar, an allowed palette or instrument set, and held-out combinations of motifs. Structural novelty is computed on symbolic features and local repetition statistics; perceptual similarity is computed with a representation fixed before the test. Raters score novelty, coherence, and usefulness independently under blind and source-labeled conditions. The study reports the label effect, the rater reliability, and the percentage of artifacts that satisfy the explicit grammar. A model that exploits an evaluator’s stylistic bias without maintaining structural diversity is identified as evaluator optimization.

D. Code and algorithms

Code tasks provide unusually strong value functions. The task card includes a hidden test suite, an execution time limit, a memory limit, and a license-safe input domain. Novelty is measured on abstract syntax and behavior, not only token distance. Usefulness is the hidden-test score and resource efficiency. Process creativity is tested by changing function names, input ordering, or an algorithmic constraint and asking for a revision. The evaluator keeps failed candidates, compiler messages, patches, and reverted patches. It therefore distinguishes an adaptive debugging loop from a single memorized solution.

E. Scientific hypotheses

For scientific hypothesis generation, the artifact is not treated as a discovery. A task card describes a bounded mechanism space and requires a proposed mechanism, predicted observations, competing explanations, and at least one falsification test. Domain experts may assess clarity and plausibility in a low-stakes blinded review, but the primary process criterion is internal testability. Held-out transfer changes the domain vocabulary while preserving the causal pattern. A generated hypothesis is never published or used to guide a real experiment without independent human review.

F. Cross-domain recombination

Cross-domain tasks directly test the combination operation suggested by computational creativity theories. Two source concepts are held out from the retrieval index and represented by controlled descriptions. The system must produce a third concept that uses a declared relation from each source but does not copy phrases, diagrams, or parameters. A retrieval-matched recombination baseline receives the same source descriptions but no search or self-critique. Value is measured by a task-specific feasibility test, and novelty is evaluated against the union of both source domains. This condition is a stress test for transfer, not a claim that combination alone is human creativity.

TABLE III
ABLATION QUESTIONS AND DIAGNOSTIC OBSERVATIONS

Removed component	Diagnostic question
Retrieval	Does novelty remain after reference access is removed and leakage is tested?
Search	Does candidate breadth improve value, or only increase evaluator-facing style?
Self-critique	Does revision correct objective errors or collapse diversity?
Human feedback	Which gains come from the person rather than the model?
Held-out split	Does the mechanism transfer beyond familiar concepts?
Provenance	Can attribution be reconstructed when the artifact is identical?

IX. ABLATIONS AND CAUSAL ATTRIBUTION

The factorial design is useful only if it can identify mechanisms. A staged ablation therefore removes one capability at a time while holding the model, budget, prompt length, task instance, and random-seed schedule fixed. The baseline emits one sample without retrieval, self-critique, or iterative human input. The generate-and-rank condition adds candidate comparison; the tree-search condition adds explicit branching; the retrieval condition adds a frozen reference index; and the critique condition adds revision feedback. The observed difference is interpreted conditionally, not as a universal contribution of “AI creativity.”

For human-AI interaction, contribution is logged at operation level. An operation is a prompt, candidate acceptance, rejection, edit, parameter change, critique, or veto. The log assigns source as human, AI, or shared and stores a cryptographic content identifier for the affected artifact. The attribution statistic in (6) is reported with sensitivity analyses for operation weights. We do not interpret a high AI fraction as autonomous agency: the human may still have supplied the goal, the evaluation function, and the acceptance rule. Provenance describes causal workflow participation, not inner mental states.

The source-label experiment uses a fixed artifact set. Each artifact is shown once with an AI label, once with a human label, and once with a neutral anonymous label in a balanced design, subject to an evaluation protocol that prevents repeated-memory effects. If scores change, the change is evidence of expectancy bias. It cannot be corrected by simply averaging labels, because the direction and size of the bias may depend on task family and evaluator expertise.

X. SAMPLING, POWER, AND ROBUSTNESS

The default offline sample is at least 20 independent candidates per condition and task instance. This number is an operational minimum for observing diversity and failure modes, not a universal power guarantee. Before finalizing the sample, a pilot on development concepts estimates variance and the smallest practically important effect. The held-out test remains untouched by tuning. If a participant evaluation is added, the number of raters and artifacts is chosen from a

simulation of the preregistered hierarchical model rather than from a generic rule of thumb.

Let d denote the smallest practically important difference on a metric, s^2 its pilot variance, and ρ the estimated within-instance correlation. A simple planning quantity for a balanced two-condition comparison is

$$n_{\text{eff}} = \frac{2s^2(1 - \rho)}{d^2}, \quad (8)$$

which is then inflated for clustering, attrition, and multiplicity. Equation (8) is a planning aid only; the final analysis uses the hierarchical model and reports uncertainty intervals. No sample size is selected after viewing held-out outcomes.

Robustness is evaluated over random seeds, prompt paraphrases, reference-corpus swaps, and small constraint changes. The primary result is stable only if its direction and practical interpretation survive these perturbations. A model may show a large novelty effect with one embedding model and no effect with another; that pattern is reported as representation sensitivity, not as a decisive creativity result. Similarly, if an artifact passes a simulator only after an undocumented manual repair, the repaired version and the original version are separate observations.

Missing data are categorized before analysis. A timeout, invalid output, and evaluator refusal are not silently removed as failures of convenience. The exclusion rule is declared by task family and applied before comparing conditions. For human ratings, “insufficient information” is a valid response and is summarized rather than imputed as a low creativity score. For model traces, a missing log invalidates the corresponding process claim but need not invalidate an independently scored artifact.

XI. EXPECTED OBSERVATION PATTERNS AND FALSIFICATION

The design supports several informative patterns. High novelty and value with low transfer would support a bounded product claim but weaken a process-level claim. High transfer after retrieval removal would weaken the memorization explanation, especially if it survives task and prompt perturbations. Search may improve objective value while reducing diversity; that result would support a search-value tradeoff rather than a simple “more search is more creative” story. A human-AI advantage that disappears when the human is prevented from selecting the final artifact would reveal hidden human selection as the main mechanism.

The benchmark can also produce null results. If AI-only, human-only, and human-AI conditions are statistically indistinguishable on a task family, the family may not separate the relevant mechanisms or the task may be too easy. If novelty and usefulness are negatively correlated, the result identifies a product tradeoff. If source labels strongly alter ratings while objective tests remain fixed, ratings are not suitable as the sole criterion. If nearest-neighbor similarity remains high despite a large semantic distance score, representation choice has failed to control memorization.

TABLE IV
INTERPRETATION OF POSSIBLE PROFILES

Observed profile	Supported interpretation
High novelty, low value	Unusual generation, not product creativity
High novelty and value, low transfer	Bounded product creativity
High value and transfer, retrieval controlled	Evidence for transferable process capability
Joint advantage, complementary edits	Conditional co-creativity
Label effect without objective change	Evaluator expectancy bias
Large nearest-neighbor overlap	Retrieval or memorization concern

The strongest falsification of H1 is not a low score on one aesthetic task. It is a replicated failure across held-out concepts, objective value, and perturbation tests after retrieval-matched controls. The strongest falsification of H3 is the absence of joint advantage after time, information, and human selection budgets are matched. H0 is weakened when an AI retains novelty, value, and transfer after retrieval and recombination controls. These are conditional inferences; they do not prove or disprove a machine’s subjective experience.

XII. IMPLEMENTATION CHECKLIST AND REPORTING SCHEMA

Before execution, the study operator completes a one-page task card and freezes the configuration. The card records task family, task generator, reference corpus version, representation model, similarity threshold, value function, hard constraints, seed schedule, model version, retrieval setting, tool permissions, time budget, and stopping rule. A second reviewer may inspect the card at the milestone, but the study does not use a repeated specification-and-quality gate for every small change.

During execution, the runner writes an append-only event log. Each event includes a task identifier, condition, seed, model identifier, input hash, output hash, tool call, latency, resource metadata, evaluator version, and error category. Human interaction events additionally record the operation type and whether the person accepted, edited, rejected, or vetoed a candidate. The log is the unit of audit; screenshots or final artifacts alone are insufficient for process claims.

The final report contains a condition-by-metric matrix, per-task distributions, held-out transfer, nearest-neighbor examples, failure categories, label-bias estimates, and provenance summaries. It lists all planned and exploratory analyses separately. It states which data were private, which model weights were unavailable, and whether any manual repair occurred. It reports negative and invalid outputs, not just successful artifacts. Every figure and table carries the task split and evaluation visibility (blind or source-labeled).

The release package is organized into six layers: task specifications; public data and licenses; model adapters; raw provenance logs; metric implementations; and analysis notebooks or scripts. A manifest maps every reported number to input hashes and a code version. For private human data, release aggregate statistics and deletion procedures instead of

raw text. For copyrighted reference materials, release hashes and derived features only when redistribution is not permitted. These procedures prevent a later reader from confusing a reproducible protocol with an unverifiable demonstration.

XIII. DISCUSSION

The proposed framework treats creativity as a family of functional properties. This choice is scientifically conservative but not empty. Engineering systems routinely need novelty, value, constraint satisfaction, and adaptive search; measuring these properties can guide design tools without deciding whether a machine has a mind. The framework also accommodates human-AI systems in which the most useful unit is a team, while preserving the distinction between joint capability and machine replacement.

The word “human” remains important. Human creativity is embedded in development, language, social learning, bodily interaction, and long-term goals. A benchmark that matches a human rating on a short task cannot represent this entire history. The protocol therefore uses human-like generalization only as a stress-test analogy, not as a definition of personhood. To claim equivalence would require a much broader theory, including how goals are formed, how values are maintained, and how an agent understands its own activity. Those questions are not identifiable from a generated artifact.

The framework has practical implications for tool builders. If a system improves value by search but loses diversity, interfaces can expose multiple candidates and preserve human choice. If held-out transfer fails, deployment should remain within validated domains. If source labels dominate ratings, product demonstrations should disclose uncertainty and avoid using anthropomorphic language. If provenance is incomplete, authorship and intellectual-property claims should remain conditional. These are engineering and governance consequences of measured behavior, not judgments about whether a machine is alive.

XIV. EXTENSIONS AND BOUNDARY CONDITIONS

The first deployment of the benchmark should remain deliberately modest. A useful extension is a longitudinal protocol in which the same system receives feedback over several sessions. The task card can then distinguish one-shot novelty from accumulation of a reusable strategy. Longitudinal data must preserve model checkpoints and prompt history; otherwise a later system update can be mistaken for learning. The analysis can compare early and late revision gain while holding the task distribution fixed.

A second extension studies interactive goal formation without treating it as human-like motivation. The system may be offered several explicitly stated goals and allowed to select a subgoal according to a declared policy. The benchmark can measure whether the selection improves coverage and value under a budget. It cannot infer that the system “wanted” the goal. This boundary prevents an observable control policy from being converted into an unsupported psychological claim.

A third extension evaluates physical robots. Embodied tasks add perception noise, actuator limits, safety constraints, and material wear. The benchmark must first run in a digital twin with a bounded action space, then in a supervised laboratory sandbox. Robots must have an emergency stop and a human operator. Physical embodiment can reveal whether creativity depends on sensing and manipulation, but it cannot by itself resolve the question of subjective experience. All hardware trials should therefore report functional outcomes and safety events, not anthropomorphic interpretations.

The benchmark also needs an adversarial evaluation track. An evaluator-independent judge, a simulator, and a held-out human panel can be compared on the same artifact set. Adversarial prompts can request a superficially surprising but infeasible object, while constraint-preserving prompts can request a conservative variant. If a system changes only its rhetorical explanation while its objective performance remains fixed, that separation is evidence about presentation, not creativity. If it finds a better design but cannot explain the constraint tradeoff, the artifact may still be useful, but the process trace is incomplete.

Finally, future benchmark versions should maintain a living contamination ledger. Each new public task is checked against known training and retrieval sources, and any later release of reference material is recorded. Benchmark scores are not comparable across versions unless the task cards, representations, and split definitions are aligned. The versioned ledger is especially important for generative models whose training data may change without an obvious user-visible change. These extensions reinforce the main principle: measure what the system did, under which information and constraints, and what a human contributed. Do not infer more than the measurement identifies.

APPENDIX A MINIMAL ARTIFACT MANIFEST

Every executed condition is accompanied by a machine-readable manifest with the following fields: benchmark version, task-family identifier, task-instance hash, development or held-out split, model and adapter version, retrieval index hash, prompt template hash, seed, sampling parameters, tool permissions, time budget, objective-function version, hard-constraint version, output hash, and evaluator version. The manifest also records whether a human selected or edited the final artifact. A missing field blocks the associated process or attribution claim until it is repaired.

The minimal event schema contains `event_id`, `parent_event_id`, `source`, `operation`, `input_hash`, `output_hash`, `timestamp`, `condition`, `task_id`, and `error_class`. For a search trace, parent identifiers reconstruct the candidate tree. For a co-creation trace, `source` is one of human, AI, or shared, and `operation` is one of propose, critique, accept, reject, edit, veto, or revise. The schema avoids storing private participant text in the public release; a salted hash and an aggregate operation

record are sufficient for audit when raw data cannot be shared.

The minimal result table has one row per artifact and columns for novelty, usefulness, surprise, diversity contribution, constraint satisfaction, held-out transfer status, nearest-neighbor similarity, and validity. A separate process table contains search breadth, revision gain, error correction, and perturbation robustness. A separate co-creation table contains human edit distance, veto rate, accepted suggestion rate, and source-attribution statistics. This separation prevents a high product score from silently substituting for evidence of a transferable process.

The release checklist is: verify licenses; remove personal identifiers; publish the task cards; freeze the split; validate hashes; run the preregistered analysis; preserve negative and invalid outputs; report all exclusions; and state whether the final artifact was human-selected. A report that omits these fields may still describe a demonstration, but it is not a reproducible creativity claim under this benchmark.

APPENDIX B CONCLUSION

Robots and AIs can plausibly produce artifacts that are novel, useful, and surprising in bounded domains. The available evidence, including AlphaGo, supports that functional claim but does not prove that a machine has human creativity. The proposed CREATIVE-AI-AGENCY-BENCHMARK turns the larger question into a sequence of falsifiable tests: control retrieval, score value and novelty separately, test held-out transfer, log the search process, and measure human contribution. A positive result would justify a bounded claim about creative performance or co-creation. A negative transfer or memorization control would identify the boundary of that claim. This is a more useful scientific outcome than declaring a machine creative or non-creative from a memorable single output.

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